

Knuth - Morris - Pratt (KMP) Algorithm

pattern: abcdabc
1 2 3 4 5 6 7

prefix: a, ab, abc, abcd,

Suffix: c, bc, abc, dabc,

C this is matching so this is the prefix
that is also there in suffix



Knuth-Morris-Pratt (KMP) Algorithm

$\bar{A}$ or lps

P_1 : $\overset{1}{a}\overset{2}{b}c d \overset{1}{a}\overset{2}{b}e \overset{1}{a}\overset{2}{b}f$
0 0 0 0 1 2 0 1 2 0

P_2 : a b c d e $\overset{1}{a}\overset{2}{b}f$ $\overset{1}{a}\overset{2}{b}c$
0 0 0 0 0 1 2 0 1 2 3

P_3 : a a b c a d a a b e
0 1 0 0 1 0 1 2 0 0

P_4 : a a a a b a a c d
0 1 2 3 0 1 2 0 0

here the next day is also same so 2 and
rest of all the values are zeros so



Knuth - Morris - Pratt (KMP) Algorithm

String: a b a b c a b c a b a b a b dⁱ
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

pattern: ⁰

	¹	²	³	⁴	⁵ ^j
a	b	a	b	d	
0	0	1	2	0	

a pattern so there is a pattern
available inside

